Activity 1E: Using a Photoresistor to Operate an LED

Purpose: To include a photoresistor that uses light detection to operate an LED.

Materials: Computer, Arduino Uno Board, USB cable, Breadboard, 1 LED, 1 - 220 Ω Resistor, 1 - 10KΩ resistor, 7 Jumper Wires, 1 Photoresistor

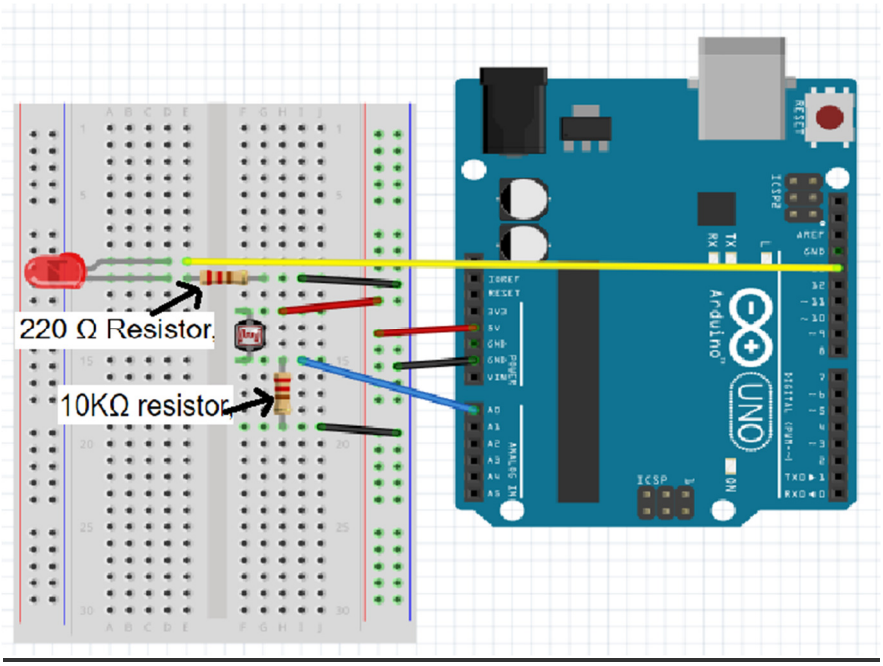
New Components:

Photoresistor. Photoresistors are light sensitive, variable resistors. As more light shines on the sensor's head, the resistance between its two terminals decreases. Therefore photoresistors can act as an switch that uses light detection to turn loads (such as light) on and off.



Procedure:

1. Wire the project using the following schematic.



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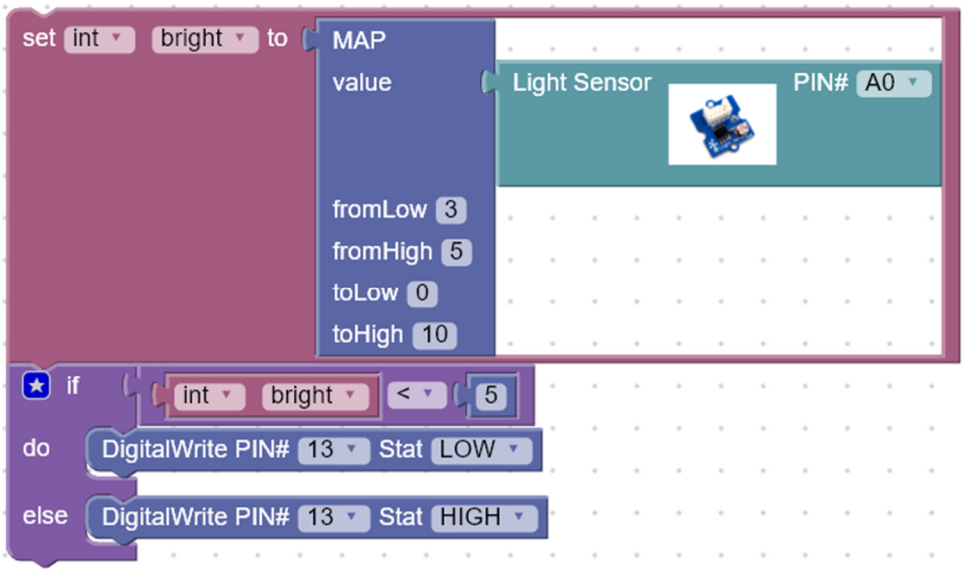
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2. You can create your code in the BlocklyDuino Editor then paste it to the Codebender website. Click BlocklyDuino Reference for a link to all the blockly codes.

3. Below are the blocks that you will need, click on the Arduino button to convert the blocks into Arduino code.
4.
    a. First we need set up a variable. These are similar to X and Y you use in math class. **Variables** are used to store information to be referenced and manipulated in a computer program. They also provide a way of labeling data with a descriptive name, so our programs can be understood more clearly by the reader and ourselves. It is helpful to think of variables as containers that hold information. Their sole purpose is to label and store data in memory. This data can then be used throughout your program. We will create the integer variable "bright".

    b. We are also going to use the "**map**" function (FIND in the MATH blocks). It re-maps a number from one range to another. That is, a value of fromLow would get mapped to toLow, a value of fromHigh to toHigh, values in-between to values in-between, etc.



*****SAVE your sketch to your BlocklyDuino Folder that you created*****

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Now select the “Arduino” tab you will see the following code.

Blocks | Arduino

```
int bright;

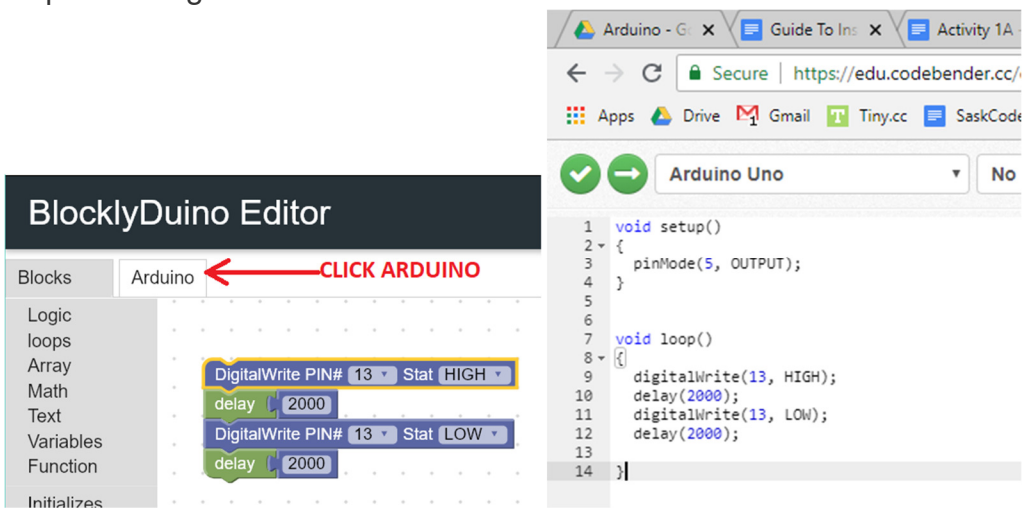
void setup()
{
  pinMode(13, OUTPUT);
}


void loop()
{
  bright = (map(round((float)(1023-analogRead(A0))*10/analogRead(A0)),3,5,0,10));
  if (bright < 5) {
    digitalWrite(13, LOW);

  } else {
    digitalWrite(13, HIGH);

  }
}
```

6. Copy the Arduino Code from BlocklyDuino into the [Codebender website](https://codebender.cc) To paste the code into the website be sure to use ctrl “v.” Hook up your arduino to your computer using the USB cable.



6. Upload your program to the arduino. Make sure the Arduino Uno board is hooked up using the supplied USB cord to your computer and click the green “upload” button.

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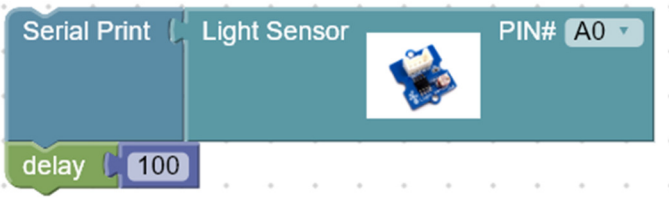
7. Your program is now loaded. Observe your LED and complete your worksheet titled **Assignment 1E: Using a Photoresistor.**

Troubleshooting: If your circuit is not working!

1. Double check your code! Did you put in every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit!
    a. Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
    b. Next check each component. Check the led, is the long positive arm connected by wires to the pins?
    c. It may be wise to test the led by turning them it on. Open a new blocklyduino window and create this test code. The block should look like this:



    d. LED still not working? Next inserting the LED directly into pin 13 and the adjacent ground. Does it light when you try this test code?
    e. Next check the wires in the circuit. This can be done by connecting the LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.
    f. You can also check the photoresistor by checking what signal is being sent to the pin using the serial monitor. With code similar to this:



    Do the values change when you cover the light sensor?
    g. For further information on using the serial monitor see Activity 4A Using a Serial Monitor Display.

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Assignment 1E: Using a Photoresistor.

1. What does the LED do when you block the light to the photoresistor? Why?

2. What is the purpose a photoresistor? List some real life applications where a photoresistor is used or an application where one would be useful.

Challenges:

3. Using your coding experience from the previous challenges, make your LED flash at 2 second intervals when it is dark. Test your code by uploading it to your arduino and observing.

4. Go back to your BlocklyDuino and change your code so that your light flashes when there is light and off when there is no light. Test your code by uploading it to your arduino and observing.

5. What are some real life applications where photoresistors could be used?

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